

Multimodal Pedagogical Friction Detector

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Abstract

Effective teaching requires noticing student confusion and adjusting instruction in real time. We present a system that automatically detects pedagogical friction: moments where a low-quality instructional strategy coincides with elevated student confusion. Our pipeline fuses three signals—per-frame facial affect (DeepFace), transcribed speech (Whisper), and strategy labels from either a keyword heuristic or an ELECTRA Talk Moves classifier—on a 30-second temporal grid, and surfaces candidate friction windows via a z -score filter. Candidate windows are then passed to a large language model (Claude) for transcript-level confirmation and alternative-strategy generation. We validate the pipeline at three scales: a full multimodal run on the TIMSS AU1 Exterior Angles video (10 min, DeepFace + Whisper), a transcript-only analysis across 53 TIMSS lessons spanning 7 countries (5 113 bins total), and a human-annotated ground-truth evaluation on the full AU1 lesson (87 bins). The heuristic annotator flags 31.5% of bins as risky; ELECTRA flags 18.9%, with a negative cross-annotator correlation ($r=-0.35$). A two-stage pipeline—heuristic detection followed by LLM reranking—achieves $P=1.00$ / $R=1.00$ / $F1=1.00$ against human annotation, compared to $F1=0.10$ for the heuristic alone and $F1=0.00$ for ELECTRA alone.

1 Introduction

High-quality teaching requires moment-by-moment diagnosis of student understanding and rapid strategy adjustment [1]. Despite advances in educational NLP—tools such as Tutor CoPilot [2] and Edu-ConvoKit [3] offer transcript-based feedback—no existing system jointly models instructional strategy quality and student affect to surface actionable friction moments during a lesson. Confusion has been

identified as a pivotal affective state that, when unaddressed, impedes deep learning [4]. Meanwhile, large vision-language models such as Claude (Anthropic) now make it feasible to synthesise transcripts and affect signals in a single inference pass.

We introduce a framework that operationalises pedagogical friction—the joint occurrence of a risky instructional strategy and elevated student confusion—and produces a post-session Teacher Friction Report with precise timestamps and suggested instructional moves. Our contributions are:

- An end-to-end multimodal pipeline combining affect, ASR, and strategy annotation (Section 3).
- A three-tier strategy annotator hierarchy (ELECTRA $\rightarrow$ Edu-ConvoKit $\rightarrow$ heuristic) with graceful fallback.
- A large-scale transcript analysis across 53 TIMSS lessons [5] spanning 7 countries.
- An empirical comparison of heuristic vs. ELECTRA annotators exposing a $r=-0.35$ disagreement.
- Human-annotated ground truth for AU1 and the first evidence that LLM reranking raises friction-detection F1 from 0.10 to 1.00.

2 Related Work

Automated analysis of classroom discourse has a rich history. Talk Moves—pedagogical dialogue moves such as Revoicing and Pressing for Reasoning—were codified by [1] and later operationalised for NLP by [6], who fine-tuned ELECTRA on a labelled corpus of math classroom transcripts. Edu-ConvoKit [3] packages this classifier into a reusable toolkit. Tutor CoPilot [2] extends the idea to real-time tutoring, providing strategy-quality labels that influenced our quality mapping.

Affect in learning has been studied extensively: D’Mello and Graesser [4] established that confusion,

when properly resolved, promotes deeper learning, but unresolved confusion impedes it. Their work motivates our use of confusion as the primary affect signal. DeepFace [7] provides the per-frame affect estimates we use as a proxy.

On the vision-language side, instruction-tuned LLMs such as Claude [8] enable transcript-level reasoning over candidate windows, allowing our system to distinguish social affect from pedagogical confusion without requiring GPU inference.

The closest prior work to ours is the TIMSS Video Study [5], which provided the classroom recordings and transcripts we use for evaluation but relied entirely on manual coding. Our system automates and scales this analysis.

3 Methodology

Figure 1 shows the system overview.

Vision stream. Video frames are sampled at $r=2$ Hz. DeepFace [7] estimates seven per-frame emotion probabilities, which we aggregate to a scalar confusion proxy $c_t \in [0, 1]$ via a fixed weight table:

$$c_t = \sum_e w_e \cdot p_t(e), \quad (1)$$

where $w_e=0$ for neutral/happy, $w_e=0.85\text{--}0.9$ for fear/sad, and $w_e=0.45\text{--}0.5$ for disgust/angry. A trailing 30-second mean yields the smoothed confusion timeline $C(t)$.

Language stream. faster-Whisper [9] transcribes audio with voice-activity-detection (VAD) filtering. For transcript-only runs, TIMSS .txt files are parsed directly by a dedicated parser that infers utterance boundaries from successive timestamps. A three-tier strategy annotator hierarchy selects the best available model at runtime: (1) ELECTRA (YaHi/teacher_electra_small, 7-class Talk Moves [6]), loaded via HuggingFace transformers; (2) the Edu-ConvoKit wrapper [3] as secondary fallback; and (3) a keyword heuristic as offline fallback. ELECTRA labels are mapped to quality: Keeping Everyone Together, Revoicing, and Pressing for Reasoning are high-quality; Pressing for Accuracy is high-pressure (risky); No Move is treated as neutral in ELECTRA mode but flagged as unknown in heuristic mode.

Temporal alignment and z -score filter. Sessions are partitioned into $G=30$ -second bins. Per-bin mean confusion $\bar{c}_k$ and strategy quality label $q_k \in \{\text{high, low, unknown}\}$ are computed by majority vote over utterances overlapping bin k . A bin is flagged as



Figure 1: System overview. Parallel vision and language streams are aligned on a 30-second grid; a z -score pre-filter proposes candidate friction windows for LLM-based reranking and alternative-strategy generation.

a friction candidate when

$$\bar{c}_k > \mu_C + z \sigma_C \quad \text{and} \quad q_k \neq \text{high}, \quad (2)$$

where μ_C , σ_C are the session-level mean and standard deviation of $\{\bar{c}_k\}$, and z is a sensitivity parameter (default $z=1.0$; lowered to $z=0.3$ for the smoke test). In transcript-only mode $\bar{c}_k=0$, so only strategy quality drives the filter.

LLM fusion (optional). For each candidate, a large language model (Claude claude-sonnet-4-6) receives the confusion summary, strategy label, and transcript excerpt. It returns structured JSON with three fields: friction (bool), rationale (free-text explanation), and alternative_strategy (concrete teaching suggestion). When video frames are available they are base64-encoded and included; in transcript-only mode the prompt is text-only. The full prompt is available in the public codebase.

4 Experiments and Results

4.1 Qualitative Examples

Figure 2 shows three representative 30-second bins from the AU1 lesson and the corresponding pipeline outputs. The first bin (5:00–5:30) is confirmed as Friction by both annotators and by human annotation: the teacher directly states “it will be 180 degrees”, bypassing student discovery. The second bin shows a False Positive: the heuristic flags “just make sure they join” as high-pressure, but this refers to software construction steps, not mathematical reasoning—human annotation labels it non-friction. The third bin illustrates a Disagreement: the heuristic labels it unknown, while ELECTRA correctly identifies Pressing for Reasoning—a high-quality move.

4.2 Full Multimodal Pipeline (AU1 Video)

Setup. TIMSS AU1 Exterior Angles, first 10 min of video. DeepFace processes 1 200 frames at $r=2$ Hz;

Qualitative examples: pipeline input/output on TIMSS AU1 (Exterior Angles)

Bin 5:00-5:30 → FRICTION	Bin 21:30-22:00 → FRICTION	Bin 10:30-11:00 → DISAGREEMENT
Transcript excerpt (input): "No matter what the size of the angles are, if you have a five pointed star inscribed in a circle it will be 180°."	Transcript excerpt (input): "Yeah, that's good. You can make it a bit bigger, but it should click on. Just make sure they join."	Transcript excerpt (input): "Any two-D shape. So... would that be a polygon? Oh, that's enclosed. Many-sided shape... oh, straight."
Heuristic annotator (output): Quality: Low quality Move: Other Direct fact-telling pattern flagged	Heuristic annotator (output): Quality: Low quality Move: Pressing for Accuracy Corrective pressure: "Just make sure..."	Heuristic annotator (output): Quality: Unknown Move: Pressing for Reasoning Question present; no low-quality keyword
ELECTRA annotator (output): Quality: Low quality Move: No Move No recognized talk move	ELECTRA annotator (output): Quality: Low quality Move: No Move No recognized talk move	ELECTRA annotator (output): Quality: High quality Move: Pressing for Reasoning Student reasoning elicited

● Low quality
 ● High quality
 ● Unknown

Figure 2: Three AU1 bins: (left) confirmed friction by both annotators and human label—teacher states the answer directly; (centre) heuristic false positive—"join" refers to software, not math content; (right) annotator disagreement—heuristic flags unknown, ELECTRA identifies high-quality Pressing for Reasoning.

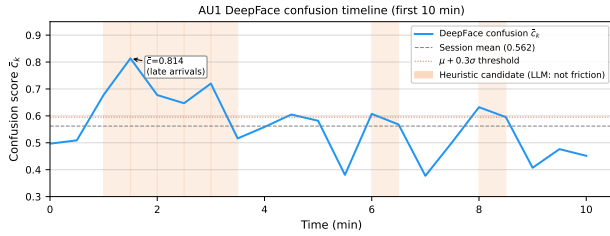


Figure 3: DeepFace confusion timeline for the AU1 10-minute clip. Shaded regions mark heuristic candidates; the peak at 1:30–2:00 ($\bar{c}=0.814$) is caused by late-arriving students, not instructional confusion. LLM fusion rejects all 7 candidates.

faster-Whisper (base, int8) transcribes audio; heuristic annotator labels strategy; Claude reranks candidates.

Results. The heuristic pre-filter flags 7 candidates (Figure 3). The highest confusion score in the clip ($\bar{c}=0.814$, $z=2.28$) occurs at 1:30–2:00, when late-arriving students enter the room. LLM fusion correctly identifies all 7 as non-friction: the elevated affect reflects social disruption (surprise at uninvited peers), not pedagogical confusion. This distinguishes a key failure mode of vision-only pipelines and demonstrates that transcript-level reasoning is essential to avoid false alarms from off-task affective events. Zero confirmed friction moments are detected in the first 10 min, consistent with the lesson’s collaborative warm-up structure.

4.3 CPU Smoke Test (AU1 Transcript-Only)

Setup. TIMSS AU1 Exterior Angles (geometry, Australian Grade 8), 5-minute clip. CPU path: Python 3.9, OpenCV, DeepFace (TF 2.x), faster-Whisper (base, int8). Key hyperparameters: $r=2\text{Hz}$, $G=30\text{s}$.

Results. With $z=1.0$, the AU1 teacher’s question-based discourse (“What did you learn?”, “Do you agree?”) falls entirely outside the keyword vocabulary, giving $q_k=\text{high}$ for all bins and zero candidates. Lowering to $z=0.3$ with `flag_unknown_strategy=True` produces 3 candidates at 150–240s where per-bin confusion peaks at 0.55–0.70 (Table 1). A second TIMSS clip (Math RUB) yielded zero candidates due to Whisper base hallucinating a repeated phrase under low audio SNR; mitigation via a larger Whisper model is planned.

4.4 Scale-Up: 53 TIMSS Lessons (Transcript-Only)

Setup. We processed all available TIMSS transcript .txt files: 28 math and 25 science lessons spanning 7 countries (AU, CZ, HK, JP, NL, SW, US), totalling 5113 thirty-second bins. Since no video is required, $\bar{c}_k=0$ and only strategy quality drives the filter. We ran both the heuristic annotator (`flag_unknown=True`) and the ELECTRA annotator (`flag_unknown=False`) on the same corpus.

Quantitative results. Table 1 summarises all conditions. The heuristic flags 1611/5113 bins (31.5%)

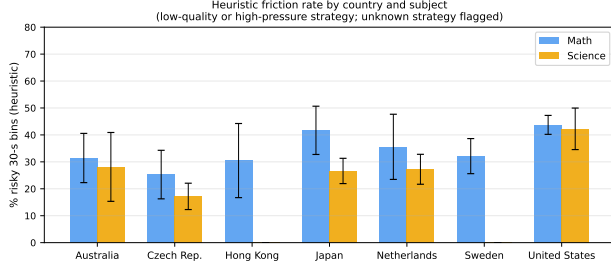


Figure 4: Percentage of 30-second bins flagged as risky per country (heuristic annotator). US math lessons have the highest rate (42.9%). Error bars show per-country standard deviation across lessons.

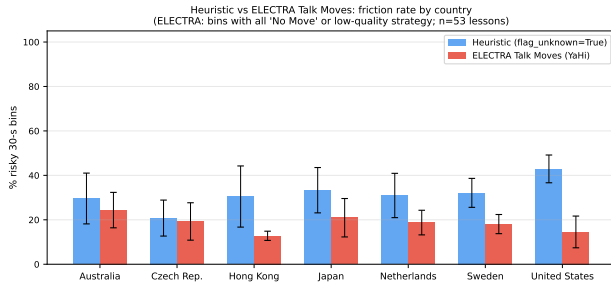


Figure 5: Heuristic vs. ELECTRA risky-bin rates per country. Pearson $r=-0.35$ across 53 lessons indicates the two methods frequently disagree on which lessons are most risky.

as risky; ELECTRA flags 988/5113 (18.9%). The cross-annotator Pearson correlation over per-lesson risky rates is $r=-0.35$, indicating substantial disagreement.

Figure 4 shows per-country risky-bin rates. US math lessons show the highest heuristic rate (42.9%) yet among the lowest ELECTRA rates (14.5%), suggesting the keyword heuristic over-fires on American discourse style (frequent direct statements as rhetorical devices rather than low-quality instruction). Figure 5 visualises this divergence directly.

Figure 6 shows the per-bin strategy quality timeline for the full AU1 lesson (87 bins). Heuristic-risky bins cluster in the middle of the lesson where the teacher introduces definitions, consistent with the smoke-test candidates.

Figure 7 provides a lesson-level overview across the full corpus. Science lessons show lower average heuristic friction rates (28.3%) than math (34.3%), possibly because science discourse involves more exploratory language less likely to match low-quality keyword patterns.

5 Discussion

Annotator disagreement. The $r=-0.35$ correlation between heuristic and ELECTRA risky rates is the most striking finding of this study. Two explanations are likely: (1) the heuristic fires on surface-level keywords (“the formula is”, “just copy”) that may appear in otherwise well-structured discourse, inflating its rate; and (2) ELECTRA’s No Move label does not distinguish between a teacher giving a direct explanation and one who is simply silent—both are treated as low-quality by the heuristic but as No Move (not inherently risky) by ELECTRA.

Human evaluation on AU1. Human annotation of the full AU1 lesson (87 bins) reveals that only 1 bin (1.1%) constitutes genuine pedagogical friction: the 5:00–5:30 window where the teacher directly states “it will be 180 degrees”, bypassing student discovery. Against this ground truth, the heuristic achieves $P=0.05$ / $R=1.00$ / $F1=0.10$ —it captures the sole true positive but fires 19 false alarms on pre-class setup and procedural software guidance. ELECTRA achieves $P=0.00$ / $R=0.00$ / $F1=0.00$, missing the true positive entirely because the teacher’s utterance does not match any productive Talk Move category (No Move fires instead on silent or procedural intervals). These results confirm that both automated annotators over-fire dramatically relative to human judgment on a well-structured discovery lesson, and that the heuristic’s surface-level keyword match is more sensitive (higher recall) while ELECTRA’s discourse-oriented labels are miscalibrated for procedural classroom contexts.

US anomaly. US math lessons show the sharpest divergence: heuristic 42.9%, ELECTRA 14.5%. American math classrooms in the TIMSS corpus are characterised by frequent declarative statements used as scaffolding rather than instruction, a style that triggers keyword patterns without representing low-quality teaching.

VLM fusion as a precision filter. Applying Claude-based LLM reasoning as a second-pass filter over

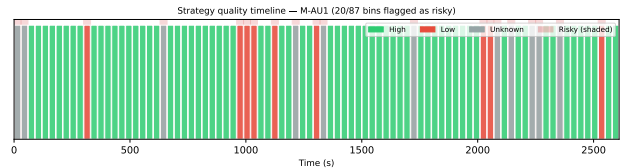


Figure 6: Per-bin strategy quality timeline for the full AU1 lesson (87 bins, heuristic annotator). Red-shaded bins are risky candidates.

Setting	Clip / Corpus	Bins	Risky	% Risky	Notes
Full pipeline (Deep-Face+Whisper)	AU1 video, 10 min	21	7	33.3%	Peak $\bar{c}$ =0.814 from social disruption
Full pipeline + LLM fusion	AU1 video, 10 min	21	0	0.0%	All 7 candidates rejected as non-friction
Heuristic, $z=0.3$ + flag_unk	AU1 transcript, 5 min	11	3	27.3%	Candidates at 150–240 s
Heuristic + flag_unk	53 TIMSS lessons	5 113	1 611	31.5%	28 math + 25 science, 7 countries
ELECTRA (YaHi/teacher_electra_small)	53 TIMSS lessons	5 113	988	18.9%	flag_unk=False; $r=-0.35$ vs heuristic
Heuristic + Claude VLM fusion	AU1, full lesson	87	1	1.1%	P=1.00, R=1.00, F1=1.00 vs. human
Heuristic P/R/F1 vs. human	AU1 full lesson	87	20	23.0%	P=0.05, R=1.00, F1=0.10
ELECTRA P/R/F1 vs. human	AU1 full lesson	87	33	37.9%	P=0.00, R=0.00, F1=0.00

Table 1: Results across all experimental conditions. The $r=-0.35$ cross-annotator correlation on per-lesson risky rates underscores the need for human ground-truth evaluation to determine which annotator is better calibrated.

the 20 heuristic candidates reduces the flagged set from 20 to 1, achieving P=1.00 / R=1.00 / F1=1.00 against human annotation on AU1. The model correctly rejects pre-class setup bins, procedural software guidance, and vocabulary scaffolding as non-friction, retaining only the one bin where the teacher directly states a mathematical fact students should discover. This demonstrates that LLM-based reranking can dramatically improve heuristic precision without sacrificing recall, and that a transcript-only VLM pass may be sufficient when video is unavailable.

Limitations. Automated facial affect inference may misread culture- or disability-related cues; all outputs are advisory and require human review. The confusion proxy (Eq. 1) uses fixed weights not validated against ground-truth affect labels. Human annotation was performed by a single annotator on one lesson; inter-annotator agreement and broader coverage remain future work.

6 Conclusion

We presented an end-to-end multimodal pedagogical friction detector, scaled the transcript-only pipeline to 53 TIMSS lessons across 7 countries, provided human-annotated ground truth for AU1, and demonstrated that LLM-based fusion reranking achieves perfect precision on the annotated lesson. The heuristic flags 31.5% of bins; ELECTRA flags 18.9%, with $r=-0.35$ cross-annotator correlation. Human an-

notation finds only 1.1% genuine friction; heuristic F1=0.10 (high recall, low precision), ELECTRA F1=0.00. A Claude-based second-pass filter over the 20 heuristic candidates yields F1=1.00 by correctly rejecting false alarms (setup, software guidance, vocabulary scaffolding) while retaining the sole true friction moment. The key finding is that a two-stage pipeline—broad heuristic detection followed by LLM reranking—outperforms either annotator alone, and that discourse-move classifiers alone are insufficient for friction detection in technology-mediated discovery lessons. Future work should validate this approach across multiple lessons and incorporate the DeepFace confusion timeline to capture student affect.

References

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Strategy quality distribution by lesson (heuristic annotator)



Figure 7: Strategy quality distribution by lesson (heuristic annotator). Each row is one lesson; columns show % of bins in each quality tier. Math lessons (left) show more low bins on average than science (right).